

STEEL MEMBER DESIGN (AS4100)

In accordance with AS4100-1998 incorporating Amendment No.1 2012

TEDDS calculation version 3.0.10

Section details

Section type

75x75x3.5 SHS C350 (BHP 1999)

Steel grade

C350

From table 2.1: Strengths of steels

Thickness of material

t = 3.5 mm

Yield stress

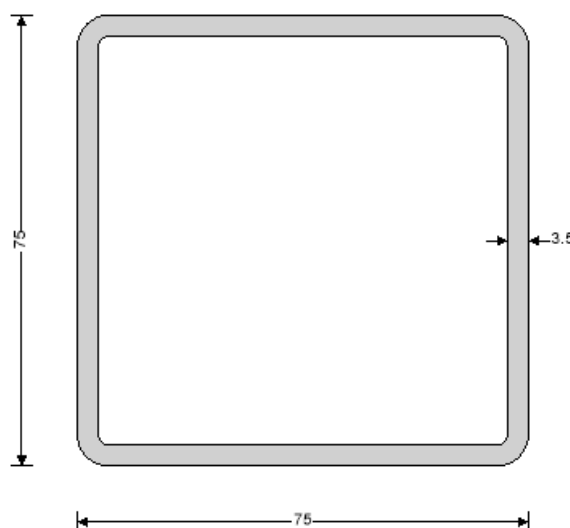
$f_y = 350 \text{ N/mm}^2$

Tensile strength

$f_u = 430 \text{ N/mm}^2$

Modulus of elasticity

E = 200000 N/mm²



Capacity factors (ϕ) for strength limit states - Table 3.4

Capacity factor

$\phi = 0.90$

Lateral restraint

Distance between major axis restraints

$L_x = 3050 \text{ mm}$

Distance between minor axis restraints

$L_y = 3050 \text{ mm}$

Effective length factors

Effective length factor in major axis

$k_{ex} = 1.000$

Effective length factor in minor axis

$k_{ey} = 1.000$

Section slenderness - Section 5.2.2

Flange slenderness

$\lambda_{ef} = (b - 2 \times t) / t \times \sqrt{[f_y / 250 \text{ N/mm}^2]} = 23.0$

Flange yield slenderness limit - Table 5.2

$\lambda_{eyf} = 40$

$\lambda_{ef} / \lambda_{eyf} = 0.575$

Web slenderness

$\lambda_{ew} = (d - 2 \times t) / (t) \times \sqrt{[f_y / 250 \text{ N/mm}^2]} = 23.0$

Web yield slenderness limit - Table 5.2

$\lambda_{eyw} = 115$

$\lambda_{ew} / \lambda_{eyw} = 0.200$

Section slenderness

$\lambda_s = 23.0$

Section plasticity limit - Table 5.2

$\lambda_{sp} = 30$

Yield slenderness limit - Table 5.2

$\lambda_{sy} = 40$

$\lambda_s < \lambda_{sp}$ - Section is compact

Project				Job no.	
Calcs for				Start page no./Revision	
				2	
Calcs by	Calcs date	Checked by	Checked date	Approved by	Approved date
S	2/06/2026				

Design for bending moment - Section 5.1

Design bending moment $M^* = 5 \text{ kNm}$

Section moment capacity for bending about a principal axis - Section 5.2

Effective section modulus - cl.5.2.3 $Z_e = \min(S_x, 1.5 \times Z_x) = 25300 \text{ mm}^3$

Nominal section moment capacity - cl.5.2.1 $M_s = f_y \times Z_e = 8.9 \text{ kNm}$

Design section moment capacity $M_{sc} = \phi \times M_s = 8 \text{ kNm}$

Segments with full lateral restraint - Section 5.3.2

End moment ratio $\beta_m = -1.000$

Maximum segment length - cl.5.3.2.4 $L_{s_max} = r_y \times (1800 + 1500 \times \beta_m) \times (b / d) \times (250 \text{ N/mm}^2 / f_y) = 6178 \text{ mm}$

Segment considered to have full lateral restraint

PASS - Design section moment capacity exceeds design bending moment

Members subject to axial compression - Section 6

Design compression force $N^* = 15 \text{ kN}$

Cross-sectional area of holes $A_h = 0 \text{ mm}^2$

Net area of cross-section $A_n = A_g - A_h = 959 \text{ mm}^2$

Nominal section capacity - Section 6.2

Flange yield slenderness limit - Table 6.2.4 $\lambda_{eyf} = 40$

Effective width of flanges - cl.6.2.4 $b_{ef} = \min(\lambda_{eyf} / \lambda_{ef}, 1) \times (b - 2 \times t) = 68.0 \text{ mm}$

Web yield slenderness limit - Table 6.2.4 $\lambda_{eyw} = 40$

Effective width of web - cl.6.2.4 $b_{ew} = \min(\lambda_{eyw} / \lambda_{ew}, 1) \times (d - 2 \times t) = 68.0 \text{ mm}$

Effective area of section $A_e = A_g - 2 \times [(b - 2 \times t) - b_{ef} + (d - 2 \times t) - b_{ew}] \times t = 959 \text{ mm}^2$

Form factor - cl.6.2.2 $k_f = A_e / A_g = 1.000$

Nominal section capacity in compression - cl.6.2.1 $N_s = k_f \times A_n \times f_y = 335.6 \text{ kN}$

Design section capacity in compression $N_{sc} = \phi \times N_s = 302.1 \text{ kN}$

Nominal member capacity in major (x-x) axis - Section 6.3

Effective length for buckling $l_{ex} = L_x \times k_{ex} = 3050 \text{ mm}$

Modified compression member slenderness $\lambda_{nx} = l_{ex} / r_x \times \sqrt{[k_f] \times [f_y / 250 \text{ N/mm}^2]} = 125.175$

Compression member factor $\alpha_{ax} = 2100 \times (\lambda_{nx} - 13.5) / (\lambda_{nx}^2 - 15.3 \times \lambda_{nx} + 2050) = 14.839$

Member section constant - Table 6.3.3(1) $\alpha_b = -0.5$

Slenderness ratio $\lambda_x = \lambda_{nx} + \alpha_{ax} \times \alpha_b = 117.756$

Compression member imperfection factor $\eta_x = 0.00326 \times (\lambda_x - 13.5) = 0.340$

Compression member factor $\xi_x = ((\lambda_x / 90)^2 + 1 + \eta_x) / (2 \times (\lambda_x / 90)^2) = 0.891$

Member slenderness reduction factor $\alpha_{cx} = \xi_x \times [1 - \sqrt{1 - (90 / (\xi_x \times \lambda_x))^2}] = 0.433$

Nominal member capacity in compression - cl.6.3.3 $N_{cx} = \alpha_{cx} \times N_s = 145.2 \text{ kN}$

Design member capacity in compression $N_{cxc} = \phi \times N_{cx} = 130.7 \text{ kN}$

Nominal member capacity in minor (y-y) axis - Section 6.3

Effective length for buckling $l_{ey} = L_y \times k_{ey} = 3050 \text{ mm}$

Modified compression member slenderness $\lambda_{ny} = l_{ey} / r_y \times \sqrt{[k_f] \times [f_y / 250 \text{ N/mm}^2]} = 125.175$

Compression member factor $\alpha_{ay} = 2100 \times (\lambda_{ny} - 13.5) / (\lambda_{ny}^2 - 15.3 \times \lambda_{ny} + 2050) = 14.839$

Member section constant - Table 6.3.3(1) $\alpha_b = -0.5$

Slenderness ratio $\lambda_y = \lambda_{ny} + \alpha_{ay} \times \alpha_b = 117.756$

Compression member imperfection factor $\eta_y = 0.00326 \times (\lambda_y - 13.5) = 0.340$

Compression member factor $\xi_y = ((\lambda_y / 90)^2 + 1 + \eta_y) / (2 \times (\lambda_y / 90)^2) = 0.891$

Member slenderness reduction factor

$$\alpha_{cy} = \xi_y \times [1 - \sqrt{1 - (90 / (\xi_y \times \lambda_y))^2}] = \mathbf{0.433}$$

Nominal member capacity in compression - cl.6.3.3 $N_{cy} = \alpha_{cy} \times N_s = \mathbf{145.2 \text{ kN}}$

Design member capacity in compression $N_{cyc} = \phi \times N_{cy} = \mathbf{130.7 \text{ kN}}$

PASS - Design capacity in compression exceeds design compression force

Members subject to combined actions - Section 8

Nominal section moment capacity - cl.8.3.2

$$M_r = M_s = \mathbf{8.9 \text{ kNm}}$$

Design section moment capacity

$$M_{rc} = \phi \times M_r = \mathbf{8 \text{ kNm}}$$

End moment ratio

$$\beta_m = \mathbf{-1.000}$$

Nominal member moment capacity - cl.8.4.2.2

$$M_i = \min(M_s \times [(1 - ((1 + \beta_m) / 2)^3] \times (1 - N^* / (\phi \times N_{cx})) + 1.18 \times ((1 + \beta_m) / 2)^3 \times \sqrt{(1 - N^* / (\phi \times N_{cx}))}, M_r) = \mathbf{7.8 \text{ kNm}}$$

Design member moment capacity

$$M_{ic} = \phi \times M_i = \mathbf{7.1 \text{ kNm}}$$

PASS - Combined axial and bending check is satisfied